21 $\int_1^3 \int_{\frac{1}{2}(y-1)}^{-\frac{1}{2}(y-5)} f\,dx\,dy$

23 $\frac{4}{15}(9\sqrt{3}-4\sqrt{2}-1) = 2.38176$

25 32/9

27 13/6

29 0

31 2/3

33 $(e-1)/2$

35 $\frac{2}{9}(3\sqrt{3}-2\sqrt{2})$

37 $\frac{1}{2}(e^2-1)$

39 (a) 8/3
(b) 16/3

41 (a) $\int_0^{1/2}\int_y^{1-y} f(x,y)\,dx\,dy$
$\int_0^{1/2}\int_0^{x} f(x,y)\,dy\,dx$
$\int_{1/2}^{1}\int_0^{1-x} f(x,y)\,dy\,dx$
(b) 1/8

43 15

45 $\int_{-5}^{5}\int_{-\sqrt{25-y^2}}^{\sqrt{25-y^2}}(25-x^2-y^2)\,dx\,dy$

47 $\int_0^4\int_{y-4}^{(4-y)/2}(4-2x-y)\,dx\,dy$

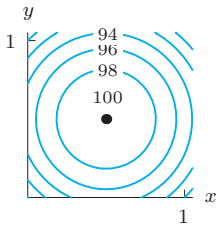
49 1/3

51 16/3

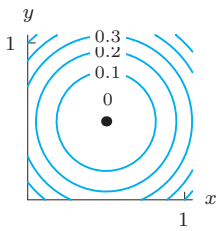
53 Volume $= 1/(6abc)$

55 (a) Circles centered at $(1,0)$
(b) $\int_{-\sqrt{3}}^{\sqrt{3}} e^{-y^2}\,dy$
(c) $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} e^{-(x-1)^2-y^2}\,dy\,dx$

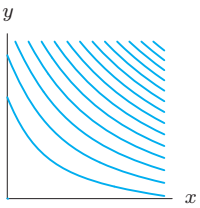
57 (a)



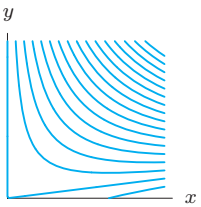
(b)



59 (a) $(4/3)a + b + (4/3)c = 20$
(b) $f(x,y) = x^2 + \frac{44}{3}xy + 3y^2$:



$f(x,y) = -3x^2 + 24xy$:



61 $k(a^3b + ab^3)/3$

63 Integrals not over same region

65 $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} 2\,dx\,dy$

\+ 67 $\int_0^2\int_0^{6-3y} 1\,dx\,dy$

69 False

71 False

73 False

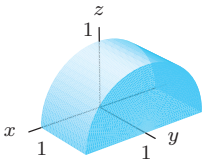
75 False

Section 16.3

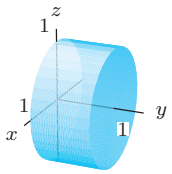
1 2

3 −8

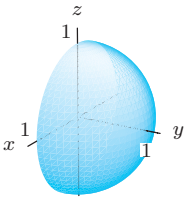
5



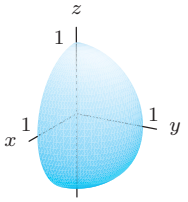
7



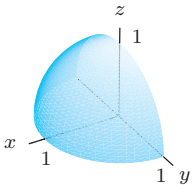
9



11



13



15 Positive

17 Positive

19 Zero

21 Positive

23 Zero

25 Positive

27 Positive

29 125

31 (a) $2\int_0^{10}\int_0^1\int_y^1 e^{-3x}\,dz\,dy\,dx$
Other answers are possible
(b) $(1-e^{-30})/3$

33 $V = \int_0^1\int_0^2\int_{x+y}^{1+2x+2y} 1\,dz\,dy\,dx$
Can reverse order x, y

35 $V = \int_0^1\int_0^{1-x}\int_{6-3x-4y}^{6-2x-2y} 1\,dz\,dy\,dx$
Can reverse order x, y

37 $V = \int_{-\sqrt{5}}^{\sqrt{5}}\int_{-\sqrt{5-x^2}}^{\sqrt{5-x^2}}\int_2^{\sqrt{9-x^2-y^2}} 1\,dz\,dy\,dx$
Can reverse order x, y

39 $\int_0^1\int_{-r}^{r}\int_0^{\sqrt{r^2-x^2}} f(x,y,z)\,dz\,dx\,dy$

41 $\int_{-r}^{r}\int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}}\int_0^{\sqrt{r^2-x^2-z^2}} f(x,y,z)\,dy\,dz\,dx$

43 1/2

45 162

47 1

49 29 gm

51 243

53 (a) $x+y+z=1$
(b) 1/6

55 $\int_0^2\int_0^{1-\frac{z}{4}}\int_0^{2-2y-\frac{z}{2}} f(x,y,z)\,dx\,dy\,dz$

57 $\int_{-4}^{4}\int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}}\int_3^{\sqrt{25-x^2-y^2}} dz\,dy\,dx$

59 $\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} dy\,dz\,dx$

61 $\int_0^2\int_0^{6y^2}\int_0^{\sqrt{12-3y^2}} f(x,y,z)\,dx\,dz\,dy$

63 4

65 $m = 2$;
$(\bar{x},\bar{y},\bar{z}) = (13/24, 13/24, 25/24)$

67 $5m/3$

71 Need same limits on innermost integral

73 $f(x,y,z) = z$

75 True

77 True

79 False

81 True

83 True

Section 16.4

1 $\int_0^{\pi/2}\int_0^{1/2} f\,r\,dr\,d\theta$

3 $\int_{\pi/4}^{3\pi/4}\int_0^2 f\,r\,dr\,d\theta$